

Comparative Analysis of Social Determinants of Health Influencing Childhood Vaccination Completion Across Indiana Counties

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Abstract: Childhood vaccination is a primary prevention of transmission of infectious diseases and pediatric morbidity. Nonetheless, there is a wide range of disparity in the administration of the recommended 4:3:1:3:3:1:4 childhood vaccine series in Indiana counties despite a consistent statewide recommendation. This project assesses the question of whether educational attainment is the most powerful social determinant of vaccination completion in comparison with median household income, accessibility of preventive services, social vulnerability, and the rural-urban classification. Descriptive statistics, regression, and machine-learning models were used to analyze and store county-level data to determine population-level associations. This work aims to support the targeted vaccination-equity actions by determining the structural obstacles that define pediatric immunization trends within the Indiana communities.

Keywords: childhood vaccination, immunization completion, social determinants of health (SDOH), education attainment, rural health, data analysis, machine learning, Indiana counties.

1 Project Scope

1.1 Introduction

In the United States, childhood vaccination is one of the most cost-effective interventions that avoids morbidity, mortality, and economic burden in the long run. Since the year 1994, routine immunizations conducted under the Vaccines for Children Program have prevented millions of hospitalizations and saved the nation billions of healthcare expenditures [10]. These results are supported by peer-reviewed evidence showing that there are substantial decreases in vaccine-preventable diseases [8] and larger developmental and economic gains throughout the lifespan [6].

Although these benefits exist, the rate of vaccination uptake differs among communities in the United States, especially among pediatric groups with socioeconomic or geographic disadvantage. It is proposed that recommended vaccine-series completion is highly dependent on structural barriers, including socioeconomic status, education of caregivers, and access to healthcare, instead of parental preference itself, as postulated by national surveillance [7]. The role of education as a determinant of health within the framework of social determinants of health (SDOH) theory is that it influences the level of health literacy, economic stability, preventive care navigation, and trust in clinical systems. These processes make education a likely predictor of the completion of the pediatric vaccine-series.

Indiana offers a good background in which to analyze these relations. A total of 92 counties adhere to the same statewide immunization guidelines but differ widely in the coverage rates. A recent systematic review found transportation barriers, shortage of providers, and inaccessibility to preventive care to be some of the key factors leading to low rates of rural immunization [1]. The disparity based on income also continues to exist; children in poorer families are more likely to be left unvaccinated because of insurance instability and limited access to primary care [5].

These findings indicate the importance of the structural predictors of county-level vaccination disparity in Indiana, including education, income, social vulnerability, the presence of preventive services, and rurality.

1.2 Aim

This paper analyzes the population-level predictors of childhood vaccination completion to the 4:3:1:3:3:1:4 immunization series in Indiana counties, which is education attainment versus median household income, Social Vulnerability Index (SVI) scores, access to preventive services, and rural-urban classification.

- **Null Hypothesis (H_0):** County-level educational attainment is not the most significant factor associated with childhood vaccination completion rates when compared to other social determinants of health.
- **Alternative Hypothesis (H_1):** County-level educational attainment has the strongest and most significant association with childhood vaccination completion rates compared to other social determinants of health.

1.3 Purpose

This project aims to determine the county-level factors that are linked with either high or low rates of childhood vaccination completion so that the public health departments and community leaders can target the areas where interventions are most required. Through education, income, preventive care use, rural-urban typology, and social vulnerability, we will seek to identify which SDOH indicators put counties at a high risk of low vaccination coverage. The information can be used to support strategic planning, resource allocation, and enhance vaccine-education efforts, which will eventually enhance health equity among children in the State of Indiana.

2 Methodology

2.1 Steps of the Project

1. Data Collection

County-level datasets were compiled from state and federal sources, including Indiana Immunization Assessment Reports, CMS SDOH data, and the USDA Rural-Urban Continuum Codes.

2. Data Extraction and Storage

Raw files were consolidated into a single master dataset containing education indicators, income, preventive services, SVI, metro/non-metro classification, and vaccination rates.

3. Data Cleaning

Using Python (pandas), we standardized variable formats, addressed missing values, validated numeric columns, and prepared the final analytic dataset for modeling.

4. Data Analysis

Descriptive statistics, correlation matrices, linear regression, logistic regression, and a random forest model were used to evaluate predictive relationships between SDOH factors and childhood vaccine completion.

5. Data Visualization

Graphs, feature-importance plots, and cluster visualizations were developed to illustrate county-level patterns and model results.

2.2 Original Team Members and Responsibilities

Team Member	Responsibilities
Adrian	Data extraction, cleaning, and merging using SQL
Alaina	Clustering and Tableau visualization
Aishwarya	Logistic regression and Random Forest modeling
Partha	Descriptive statistics and correlation analyses

2.3 Actual Contributions from Individual Team Members

The roles were re-allocated in our group as the project went on, as each person gained confidence, technical skills, and familiarity with certain tools. We did not impose a strict division of labor but instead used a collaborative organization where members assumed the major responsibility of the parts they could comfortably do. This method enhanced efficiency, minimized re-work, and enabled every member to work on a greater level of technical depth. There was a lot of cross-checking of the work of other team members, but the ownership of the tasks was correlated with the data skills, statistical familiarity, and programming confidence, which led to a more effective workflow.

Team Member	Responsibilities
Adrian	Machine Learning & Coding: · Located and downloaded relevant public datasets for model testing. · Ran Python code for machine learning (logistic regression and linear regression). · Conducted train/test splits and binning of variables. Generated model outputs (confusion matrices, ROC curves, classification reports). · Evaluated predictive performance and documented model weaknesses. · Helped troubleshoot Python environment and execution errors. · Assisted with labeling machine-learning visuals for inclusion in the report.

Alaina	Data Cleaning, Methodology, Documentation: · Located vaccination and SDOH data from Indiana health portals and CMS sources. · Performed detailed data cleaning and sorting (removing nulls/negatives, formatting percentages, verifying column types). · Wrote major portions of the methodology section, including data-preparation steps. · Led formatting of the final report, figure placement, and LNCS-style structure. · Organized citations and reference formatting to ensure academic alignment. · Verified narrative flow between sections and helped finalize wording across the paper. · Supported dataset organization and quality-control before analysis.
Aishwarya	Data Cleaning, Statistical Preparation, SQL → Python: · Assisted in downloading and preparing SDOH datasets. · Cleaned and pre-processed numeric fields to ensure valid statistical input. · Converted exported SQL outputs into Python data frames. · Ran statistical tests (normality checks, distribution plotting, preliminary comparisons). · Verified assumptions prior to model execution. · Helped prepare tables or values for figures used in the data-analysis section. · Supported citation gathering by confirming federal dataset sources.
Partha	Correlation Testing, Visualization, SQL Conversion: · Helped locate county-level data and confirm variable definitions. · Converted raw data sources into SQL-ready tables and assisted with Python imports. · Conducted correlation analysis (Spearman correlations between SDOH and vaccination). · Produced visualizations such as heatmaps and exploratory scatter plots · Identified variables with statistically significant relationships · Assisted in re-checking data accuracy alongside the cleaning workflow. · Contributed citation support for datasets used in statistical testing.

2.4 Project Challenges

Although our group worked efficiently, we had to face several analytical, technical, and logistical issues during the project. Among the most stable challenges was finding a time that could be agreed upon by all members to hold the meetings. The work, academic, and personal schedules of the team differed, and in most cases, they could only meet on Tuesdays and Thursdays, after 5:00 PM, which restricted the number of synchronous troubleshooting sessions that could be done. This is occasionally sluggish development on statistical tests or Python

debugging because some processes took more than one individual to be available to look at the error, re-run code, or check the formatting of the dataset.

One of the greatest technical issues was the instability of our statistical tests and visualization results in Python. The independent variables exhibited low variance, and most of them were not linear with vaccination completion, leading to poor model fit and regression plots that were not of much interpretative value. When we tried to create scatterplots and regression lines, the data often fell in a vertical manner on the y-axis rather than creating a significant slope. These vertical charts were occasioned by the fact that the distribution of the counts of the counties was narrow, and there was no statistical separation needed to use linear modeling. Consequently, we needed to check assumptions of normality and skewness many times and run diagnostic plots to establish that variables could be used.

Unstable numerical results of our statistical tests were also a challenge to us. Several correlation analyses gave effect sizes that were too small to predict, and the descriptive statistics indicated a small spread amongst the counties. This had to re-clean the data, re-examine scaling options, and re-run histograms and scatterplots to make sure that the dataset was arranged correctly. Model assumptions were still problematic to meet even after these steps because of aggregation effects, the small size of the sample (92 counties), and non-uniform feature behavior. These challenges delayed our development, but the iterative debugging process enhanced our mastery of Python, standing to better assess the distributional assumptions and making us aware that there are some types of data in the field of public health that are not well-served by linear regression.

3 Data Collection

3.1 Sources and Acquisition

The data on county-level childhood vaccination rates were sourced as the data provided in the Indiana Immunization Assessment Reports, where the completion rates of the 4:3:1:3:3:1:4 childhood immunization schedule were recorded in all 92 counties of Indiana [10]. The CMS Mapping SDOH Dataset and Hoosier Health and Well-Being Dataset were used to retrieve the measures of education, median household income, preventive-care utilization, and Social Vulnerability Index (SVI) values, both of which offer standardized county-level health-equity research indicators [6, 8]. The values of the rural-urban classification were determined according to the USDA Rural-Urban Continuum Codes, and each county was classified as metro or non-metro according to federal classification [7]. Following the acquisition, all sources were consolidated based on county name and county identifier to provide 92 complete county-level observations. Standardization of data formats was done before analysis to make numerical fields compatible, categorical encodings, and variable naming conventions.

3.2 Variable Identification

Each dataset row represents a single Indiana county and includes the following variable groups:

County Identification

- County - county name
- County_ID - unique numeric identifier

Education Variables

- No_Formal_Ed_Count - residents reporting no formal education
- No_HS_Diploma_Count - residents without a high school diploma
- HS_Diploma_Count - residents holding a high school diploma
- Associates_Count - residents with an associate degree
- Bachelors_Count - residents with a bachelor's degree
- Grad_School_Count - residents with a graduate or professional degree
- Attending_School_Count - residents currently enrolled in school

These categories reflect the education-attainment distributions reported in CMS-based SDOH surveillance systems.

Vaccination Outcome Variables

- Completion_Rate - completion of the full 4:3:1:3:3:1:4 series (target variable)
- 4_DTaP_Rate - DTaP vaccination series completion
- Polio_Rate - polio vaccination rate
- 1_MMR_Rate - first-dose measles–mumps–rubella vaccination rate
- 3_Hib_Rate - Haemophilus influenzae type b vaccination rate
- 3_HepB_Rate - hepatitis B vaccination rate
- 1_VAR_Rate - varicella vaccination rate
- 4_PCV_Rate - pneumococcal vaccination rate

These variables correspond to the vaccine-specific performance indicators visualized in exploratory analysis and scatter-plot diagnostics.

Socioeconomic and Preventive-Care Variables

- Average_Median_Household_Income - median household income
- AVG_Current_Preventive_Services_(65+) - preventive-service utilization among adults aged ≥65 years

Geographic Classification

- METRO - urban-county indicator (1 = metro, 0 = non-metro)
- NON_METRO - rural-county indicator (1 = non-metro, 0 = metro)

The binary coding reflects the simplified rural–urban assignment used in prior Indiana county-level assessments.

Social Vulnerability

- SVI_County_Average - average SVI score for each county

This indicator represents composite population-level vulnerability derived from federal public-health scoring frameworks.

4 Data Extraction and Storage

4.1 Data Extraction

We extracted county-level vaccination and social determinants of health (SDOH) data from publicly available datasets. The extracted variables included childhood vaccination completion rates, education attainment counts, median household income, preventive care access, urban–rural classification, and the Social Vulnerability Index (SVI). We consolidated data from multiple sources into a single structured dataset to ensure consistency and completeness for analysis.

4.2 Data Cleaning

We cleaned the extracted data using Microsoft Excel prior to analysis. We removed null, negative, and invalid values and standardized percentage fields into numerical formats. We verified column names, corrected data types, and ensured uniform measurement scales across all variables. This cleaning process improved data quality and ensured the dataset was suitable for statistical analysis and modeling.

A	B	C	D	E	F	G	H	I	J	K
County	County_ID	NO_FORMAL_ED_COUN	NO_HS_DIPLOMA_COUN	HS_DIPLOMA_COUN	ASSOCIATES_COUN	BACHELORS_COUN	GRAD_SCHOOL_COUN	ATTENDING_SCHOOL_COUN	Completion Rate for	4 Dtap RATE
Adams	1	100	524	745	0	0	0	322	50%	63%
Allen	2	4516	13241	11923	1790	1019	175	7345	58%	65%
Bartholomew	3	617	1632	2505	148	76	0	2106	54%	74%
Benton	4	0	224	278	0	0	0	0	51%	58%
Blackford	5	0	404	597	0	0	0	156	59%	64%
Boone	6	845	1714	1615	98	87	0	470	71%	77%
Brown	7	51	358	297	0	0	0	214	63%	74%
Carroll	8	111	524	313	0	0	0	175	60%	67%
Cass	9	234	1294	961	0	0	0	546	57%	68%
Clark	10	1279	3410	3128	325	102	0	3678	56%	71%
Clay	11	555	1741	1502	0	0	0	297	66%	70%
Clinton	12	693	2303	1328	0	0	0	317	64%	71%
Crawford	13	110	650	536	0	0	0	48	54%	63%
Daviess	14	491	1412	972	21	0	0	223	38%	51%
Dearborn	15	377	824	650	0	0	0	1079	53%	66%
Decatur	16	79	496	780	0	0	0	791	70%	78%
Delaware	17	409	1331	866	23	0	0	512	60%	68%
Dubuque	18	1495	4559	5592	567	303	0	3251	60%	65%
Dubuque	19	604	1208	825	0	0	0	152	68%	73%
Elkhart	20	3732	9053	9456	643	269	0	2350	58%	68%
Fayette	21	225	714	597	0	0	0	1064	71%	76%
Floyd	22	578	1396	2492	73	0	0	2095	62%	72%
Fountain	23	157	683	828	0	0	0	110	49%	54%
Franklin	24	52	293	371	0	0	0	512	70%	76%
Fulton	25	243	794	837	0	0	0	168	56%	65%
Gibson	26	783	961	1726	0	0	0	280	73%	79%
Grant	27	782	2381	3309	236	0	0	1538	60%	67%

Fig 1. Original data with null (negative) Values

A	B	C	D	E	F	G	H
COUNTY	NO_HS_DIPLOMA_COUN	HS_DIPLOMA_COUN	COMPLETION_RAT	LOCATIO	AVG_MED_INCM	AVG_PREV	AVG_SVI
Adams	524	745	0.5	0	52005.67	0.3703	0.53
Allen	13241	11923	0.58	1	56925.5	0.3685	20.37
Bartholomew	1632	2505	0.54	1	67382.06	0.4076	0.47
Benton	224	278	0.51	1	52691.33	0.3638	0.61
Blackford	404	597	0.59	0	44725.25	0.3737	0.53
Boone	1714	1615	0.71	1	95305.18	0.4450	0.21
Brown	358	297	0.63	1	66617.2	0.4086	0.36
Carroll	524	313	0.6	1	54956.86	0.4179	0.32
Cass	1294	961	0.57	0	51523.36	0.3585	0.52
Clark	3410	3128	0.56	1	59938.7	0.3544	0.4
Clay	1741	1502	0.66	1	62014.17	0.3928	0.39
Clinton	2303	1328	0.64	0	52330.12	0.3863	0.54
Crawford	650	536	0.54	0	42301.67	0.3505	0.54
Daviess	1412	972	0.38	0	54160.62	0.3440	0.62

Fig 2. Cleaned data with null (negative) values

After the data was initially cleaned by deleting null and negative values, we used a heat map to further eliminate data that could be problematic in our analysis. The heat map helped us interpret the correlation between our attributes. The outliers of the data were removed by calculating the mean and standard deviation in the Python Jupyter notebook.

4.3 Data Import

We imported the cleaned dataset into a MySQL database using phpMyAdmin for structured storage and efficient access. We created a relational table to store county-level records and verified successful data insertion through database queries. We then connected the database to a Python Jupyter Notebook using MySQL connectors, enabling seamless data retrieval for exploratory analysis, visualization, and statistical modeling.

County	County_ID	No_Formal_Ed_Count	No_HS_Diploma_Count	HS_Diploma_Count	Associates_Count	Bachelors_Count	Grad_School_Count	Attending_School_Count	Completion_Rate	Dtap_Rate	Polio_Rate
Adams	1	100	524	745	0	0	0	322	0.5	0.63	0.79
Allen	2	4516	13241	11923	1790	1019	175	7345	0.58	0.65	0.8
Bartholomew	3	617	1632	2505	148	76	0	2106	0.54	0.74	0.85
Benton	4	0	224	278	0	0	0	0	0.51	0.58	0.69
Blackford	5	0	404	597	0	0	0	156	0.59	0.64	0.83
Boone	6	845	1714	1615	98	87	0	470	0.71	0.77	0.85
Brown	7	51	358	297	0	0	0	214	0.63	0.74	0.82
Carroll	8	111	524	313	0	0	0	175	0.6	0.67	0.8
Cass	9	234	1294	961	0	0	0	546	0.57	0.68	0.81
Clark	10	1279	3410	3128	325	102	0	3678	0.56	0.71	0.81
Clay	11	555	1741	1502	0	0	0	297	0.66	0.7	0.83
Clinton	12	693	2303	1328	0	0	0	317	0.64	0.71	0.77
Crawford	13	110	650	536	0	0	0	48	0.54	0.63	0.77
Daviess	14	491	1412	972	21	0	0	223	0.38	0.51	0.77
Dearborn	15	377	824	650	0	0	0	1079	0.53	0.66	0.81
Decatur	16	79	496	780	0	0	0	791	0.7	0.78	0.89
Dekalb	17	409	1331	866	23	0	0	512	0.6	0.68	0.81
Delaware	18	1495	4559	5592	567	303	0	3251	0.6	0.65	0.77
Dubois	19	604	1208	825	0	0	0	152	0.68	0.73	0.93
Elkhart	20	3732	9053	9456	643	269	0	2350	0.58	0.68	0.82
Fayette	21	225	714	597	0	0	0	1064	0.71	0.76	0.87
Floyd	22	578	1396	2492	73	0	0	2095	0.62	0.72	0.83
Fountain	23	157	683	828	0	0	0	110	0.49	0.54	0.69
Franklin	24	52	293	371	0	0	0	512	0.7	0.76	0.89
Fulton	25	243	794	837	0	0	0	168	0.56	0.65	0.84

Fig 3. Structure of vaccination rate in phpMyAdmin

5 Data Analysis & Visualization

Python was used for data analysis, in which linear regression was employed to analyze the data.

5.1 Steps to Test the Normality of the Data

Step 1

For the analysis, we connected phpMyAdmin SQL to Python Jupyter notebook.


```

[5]: myvars = {}
    with open("aismohan-mysql-password") as myfile:
        for line in myfile:
            name, var = line.partition(":")[::2]
            myvars[name.strip()] = var.strip()

[ ]:

[6]: myvars.keys()

[6]: dict_keys(['DB username', 'DB databasename', 'DB password'])

[7]: myvars ['DB databasename']

[7]: 'aismohan_db'

[8]: import MySQLdb
    conn = MySQLdb.connect(hosts="localhost", user=myvars['DB username'], passwd=myvars['DB password'], db=myvars['DB databasename'])
    cursor = conn.cursor()

[1]: import MySQLdb
    import pandas as pd

    # Connect to MySQL
    conn = MySQLdb.connect(
        hosts="localhost",
        user=myvars["DB username"],
        passwd=myvars["DB password"],
        db=myvars["DB databasename"]
    )

    df = pd.read_sql("SELECT * FROM master_sdoh_1_csv;", conn)
    df.head()
    df.info()

```

Fig 4. Python code using pandas to create a dataframe

The image below displays the data for the independent variables. This shows the successful connection between the database and Jupyter notebook.

	County	County ID	NO FORMAL ED COUNT	NO HS DIPLOMA COUNT	HS DIPLOMA COUNT	ASSOCIATES COUNT	BACHELORS COUNT	GRAD SCHOOL COUNT	ATTENDING SCHOOL COUNT	Completion Rate for 4:3:1:3:3:1:4	...	MMR RATE	3 Hib RATE	3 Hep B RATE	1 VAR RATE	4PCV RATE	METRO
0	Adams	1	100	524	745	0	0	0	322	0.50	...	0.80	0.73	0.77	0.78	0.57	0
1	Allen	2	4516	13241	11923	1790	1019	175	7345	0.58	...	0.80	0.78	0.81	0.80	0.65	1
2	Bartholomew	3	617	1632	2505	148	76	0	2106	0.54	...	0.84	0.85	0.67	0.80	0.75	1
3	Benton	4	0	224	278	0	0	0	0	0.51	...	0.67	0.70	0.68	0.67	0.59	1
4	Blackford	5	0	404	597	0	0	0	156	0.59	...	0.80	0.78	0.85	0.79	0.63	0
5	Boone	6	845	1714	1615	98	87	0	470	0.71	...	0.83	0.82	0.82	0.83	0.76	1

Fig 5. Data for independent variables

Step 2

After a successful connection between the database and the Python Jupyter notebook, we have tested whether the data is normally distributed or not. To check this, we plotted a histogram and pairplots using python packages(the seaborn package).

Initially, we checked the distribution of data, whether it is normally distributed or not, by plotting histograms.

```

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import norm

target = "Completion Rate for 4:3:1:3:1:4"
data = df[target].dropna()

plt.figure(figsize=(8,5))
plt.hist(data, bins=12, density=True, alpha=0.6, color='skyblue', edgecolor='black')

mu, sigma = data.mean(), data.std()
x = np.linspace(data.min(), data.max(), 100)
plt.plot(x, norm.pdf(x, mu, sigma), 'r-', linewidth=2)

plt.title("Normality Check: Vaccination Completion Rate")
plt.xlabel("Completion Rate")
plt.ylabel("Density")
plt.grid(alpha=0.3)
plt.show()

```

Lambda: 2.0903793387285696
 Box-Cox p-value: 0.0803770752720351

Fig 6. Python code for Box-Cox and P value

Step 3

Normality Testing and Transformation

We assessed the distribution of the vaccination completion rate to verify whether it met the assumptions required for parametric statistical analysis. Initial exploratory analysis indicated mild right skewness in the data. To address this issue, we applied a Box–Cox transformation to improve normality.

The Box–Cox transformation produced a lambda value of 2.09, suggesting that the original distribution was moderately skewed. After transformation, the distribution closely approximated a normal shape, as evidenced by the alignment of the histogram with the theoretical normal density curve.

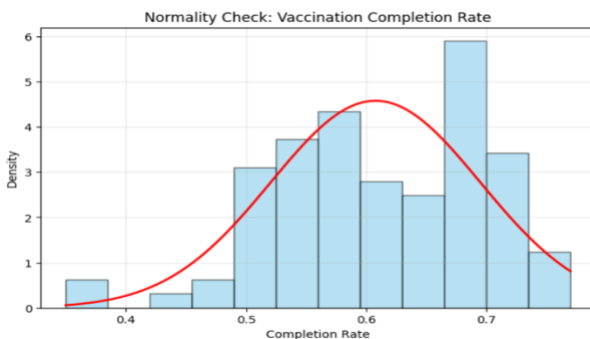


Fig 7. Normality test histogram curve

```

import numpy as np
import matplotlib.pyplot as plt

data = df['COMPLETION_RATE'].dropna()

counts, bins, patches = plt.hist(data, bins=10, edgecolor='black')

colors = plt.cm.tab10(np.linspace(0, 1, len(patches)))

for patch, color in zip(patches, colors):
    patch.set_facecolor(color)

plt.title("Distribution of Childhood Vaccination Completion Rates")
plt.xlabel("Completion Rate")
plt.ylabel("Frequency")
plt.grid(axis='y', alpha=0.4)
plt.show()

```

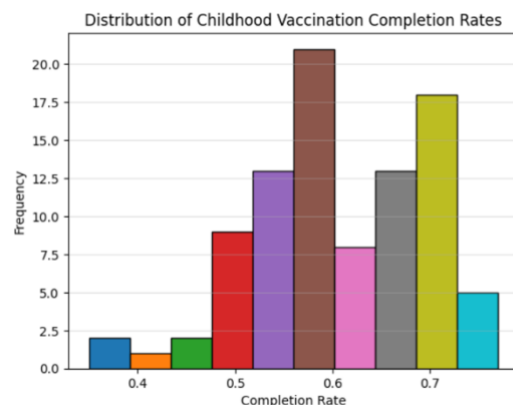


Fig 8. Python code and Distribution of Childhood Vaccination Completion Rates

The Box–Cox normality test returned a p-value of 0.080, which exceeds the 0.05 significance threshold. This result indicates that the transformed vaccination completion rate does not significantly deviate from a normal distribution. Therefore, the transformed data satisfy the normality assumption required for subsequent analyses such as linear regression, Spearman test, and exploratory analysis.

We used a histogram to examine the distribution of childhood vaccination completion rates across counties. The x-axis represents the vaccination completion rate, while the y-axis shows the frequency of counties. Most counties cluster between 0.55 and 0.70, indicating moderate to high vaccination coverage. Fewer counties appear at very low (<0.45) or very high (>0.75) completion rates, suggesting limited extreme cases. The distribution shows slight skewness, which motivated further normality testing and transformation.

Purpose:

This visualization helps identify the overall spread, central tendency, and skewness of the outcome variable before statistical modeling.



Fig 9. Python code and Distribution of Key SDOH Risk Factors vs Vaccination Completion Rate

We plotted multiple Social Determinants of Health (SDOH) indicators against vaccination completion rate. Each color represents a different SDOH variable, such as:

- No formal education count
- No high school diploma count
- High school diploma count
- Average median household income
- SVI county average

Vaccination completion rate is shown on the x-axis, while SDOH values appear on the y-axis. Counties with higher education and income values tend to align with higher vaccination completion rates. Counties with higher social vulnerability and lower education levels tend to cluster at lower completion rates.

Purpose:

This plot visually demonstrates how multiple SDOH factors relate to vaccination outcomes and highlights disparities across counties.

```
# Scatter plot
plt.scatter(x, y, alpha=0.7, s=40, color='blue')

# Trendline
if len(x) > 1:
    m, b = np.polyfit(x, y, 1)
    xs = np.linspace(x.min(), x.max(), 100)
    plt.plot(xs, m*xs + b, color='red', linewidth=2)

plt.xlabel(col)
plt.ylabel("Completion Rate")
plt.title(f"Completion Rate vs {col}")
plt.grid(alpha=0.3)
```

Fig 10. Python codes for Completion Rate vs Average Median Household Income and Average Preventive Services (65+)

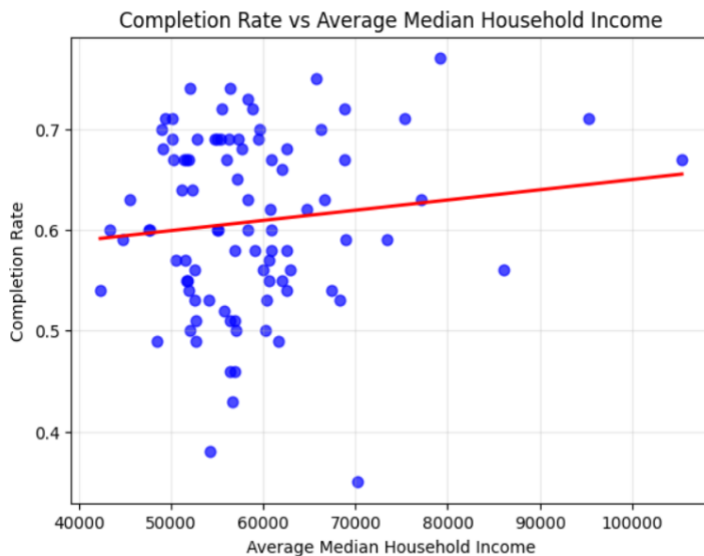


Fig 11. Completion Rate vs Average Median Household Income

We used a scatter plot to analyze the relationship between household income and vaccination completion rate. Each point represents a county. The red trendline shows a positive linear relationship. As median household income increases, vaccination completion rate generally increases. Although the relationship is not perfect, the upward trend indicates that income plays a role in vaccination access and uptake.

Purpose:

This visualization supports the hypothesis that economic stability positively influences public health outcomes.

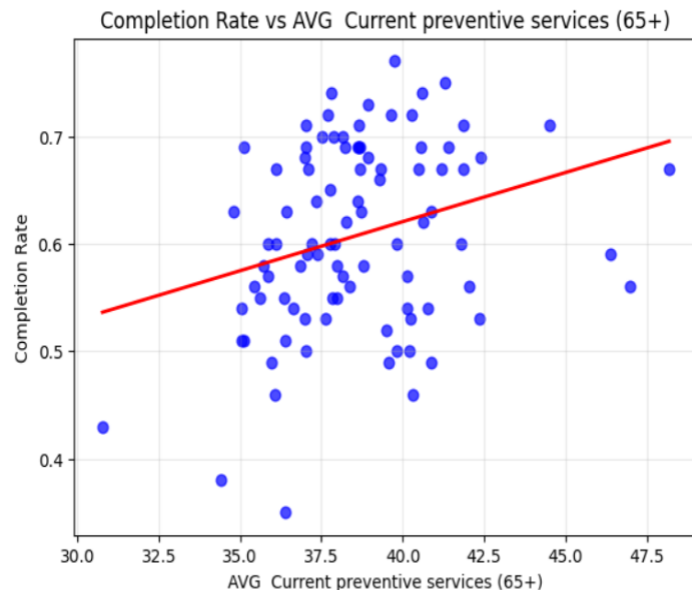


Fig 12. Completion Rate vs Average Preventive Services (65+)

We examined the relationship between access to preventive healthcare services and vaccination completion rate. The scatter plot shows a clear positive trend. Counties with higher preventive service utilization among older adults tend to have higher childhood vaccination completion rates. This suggests that a stronger healthcare infrastructure benefits broader population health outcomes.

Purpose:

This visualization highlights the importance of healthcare access as a social determinant influencing vaccination equity.

```
import pandas as pd
import matplotlib.pyplot as plt

df.columns = [c.strip() for c in df.columns]

vacc_col = "Completion Rate for 4:3:1:3:1:4"

if vacc_col not in df.columns:
    raise ValueError(f"Column '{vacc_col}' not found. Check column names.")

top10 = df_clean.sort_values(by=vacc_col, ascending=False).head(10)

plt.figure(figsize=(10,5))
plt.barh(top10['County'], top10[vacc_col], color='green')
plt.gca().invert_yaxis() # highest at the top
plt.xlabel("Vaccination Completion Rate")
plt.title("Top 10 Highest Vaccinated Counties")
plt.tight_layout()
plt.savefig("top10_vaccinated_counties.png", dpi=200)
plt.show()

print("\n ⚡ TOP 10 Highest Vaccinated Counties:")
print(top10[['County', vacc_col]])

bottom10 = df_clean.sort_values(by=vacc_col, ascending=True).head(10)

plt.figure(figsize=(10,5))
plt.barh(bottom10['County'], bottom10[vacc_col], color='red')
plt.xlabel("Vaccination Completion Rate")
plt.title("Bottom 10 Lowest Vaccinated Counties")
plt.tight_layout()
plt.savefig("bottom10_vaccinated_counties.png", dpi=200)
plt.show()

print("\n ⚡ BOTTOM 10 Lowest Vaccinated Counties:")
print(bottom10[['County', vacc_col]])
```

Fig 13. Python codes for the top and bottom 10 counties

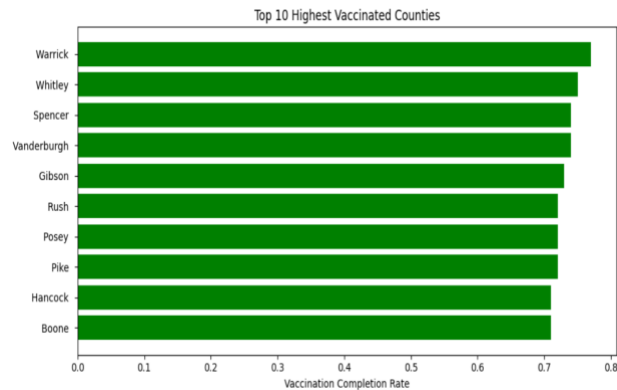


Fig 14. Top 10 Highest Vaccinated Counties

We created a horizontal bar chart to display the counties with the highest vaccination completion rates. Counties are sorted in descending order for easy comparison. These counties consistently show completion rates above the dataset average. Many of these counties also demonstrate higher income levels and better access to healthcare services.

Purpose:

This chart identifies high-performing counties that may serve as benchmarks for public health best practices

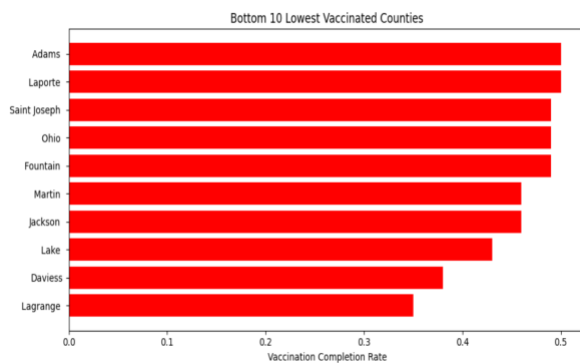


Fig 15. Bottom 10 Lowest Vaccinated Counties

We used a horizontal bar chart to highlight counties with the lowest vaccination completion rates. These counties show significantly lower coverage compared to the rest of the dataset. Many of these counties also exhibit higher social vulnerability and lower education levels. This visualization emphasizes geographic and social disparities in vaccination outcomes.

Purpose:

This chart helps identify priority regions where targeted public health interventions may be needed.

Overall, the visualizations demonstrate that childhood vaccination completion rates are not evenly distributed across counties and are strongly influenced by social, economic, and healthcare-related factors

5.2 Testing normality of the dependent variable and independent variables
Variable Identification and Data Refinement

The study designates the Completion Rate of the 4:3:1:3:3:1:4 Vaccination series as the primary dependent variable. To understand the factors influencing this rate, several independent variables were selected: geographic location (Rural vs. Urban), educational attainment (possession of a high school diploma), the Social Vulnerability Index (SVI), and Healthcare Access & Quality regarding preventative services.

To ensure the integrity of the statistical analysis, a rigorous data cleaning process was implemented. This involved distinguishing between categorical and numeric variables and identifying outliers that could skew the results. By eliminating these outliers, the dataset became more representative of general trends, allowing for a more accurate assessment of the relationships between socioeconomic factors and public health outcomes.

Determinant (Independent variable)	Type of data
Has No HS Diploma	Numeric
Has HS Diploma or Above	Numeric
Average Annual Median Household Income	Numeric
Healthcare Access & Quality (Preventative Services)	Numeric
Social Vulnerability Index	Numeric
Location	Categorical

Table 1. Determinants and their type

Output /Dependent Variable	Type of data
Completion Rate of 4:3:1:3:3:1:4 Vaccination	Numeric

Table 2. Dependent variable and its type

Normality Testing and the Shapiro-Wilk Approach

A critical step in the methodology was determining the distribution of the continuous numerical variables. The Shapiro-Wilk test was selected as the statistical tool to check for normality. This test is essential for deciding whether to use parametric or non-parametric statistical methods. As seen in the provided histograms (Fig 16.), the blue dashed lines represent the mean, and the green solid lines represent the median. When these two lines are significantly separated, or when the "bell curve" (KDE plot) appears lopsided, it indicates a departure from a normal distribution.

Columns	Type	Mean	Median	Mode
Completion Rate of 4:3:1:3:3:1:4 Vaccination	Output	0.6073	0.600	0.6700
No HS Diploma	Input	1965.7791	939.500	524.0000
Has HS Diploma or Above	Input	1821.2791	959.000	331.0000
Average Annual Median Household Income	Input	58696.74	56970.135	42301.67
Healthcare Access & Quality (Preventative Services)	Input	0.3859	0.382	0.3505
Social Vulnerability Index	Input	0.4313	0.440	0.4400

Table 3. Results of Normality Testing and the Shapiro-Wilk Approach

Observations of Skewness and Distribution Trends

The graphical outcomes confirm that the numeric variables do not follow a perfectly normal distribution. Specifically:

- **Education Variables:** The distributions for "Don't have HS Diploma" and "Has HS Diploma" show significant right-skewness (positive skew). A vast majority of the data points are clustered at the lower end, with a long tail extending toward the higher values.
- **Vaccination and SVI:** The "Completion Rate" and "Social Vulnerability Index" show more spread but remain non-normal. The Completion Rate, for instance, appears somewhat **bimodal**, with peaks near 0.6 and 0.7, suggesting distinct subgroups within the county data.
- **Income and Healthcare Access:** "Annual Median Household Income" and "Healthcare Access & Quality" also exhibit skewness, with income showing a concentration of counties in the \$50,000 to \$60,000 range and a trailing tail toward higher income levels.

Because the Shapiro-Wilk test and these visualizations highlight significant skewness, the study must account for this non-normality to ensure the findings regarding vaccination completion rates are statistically sound.

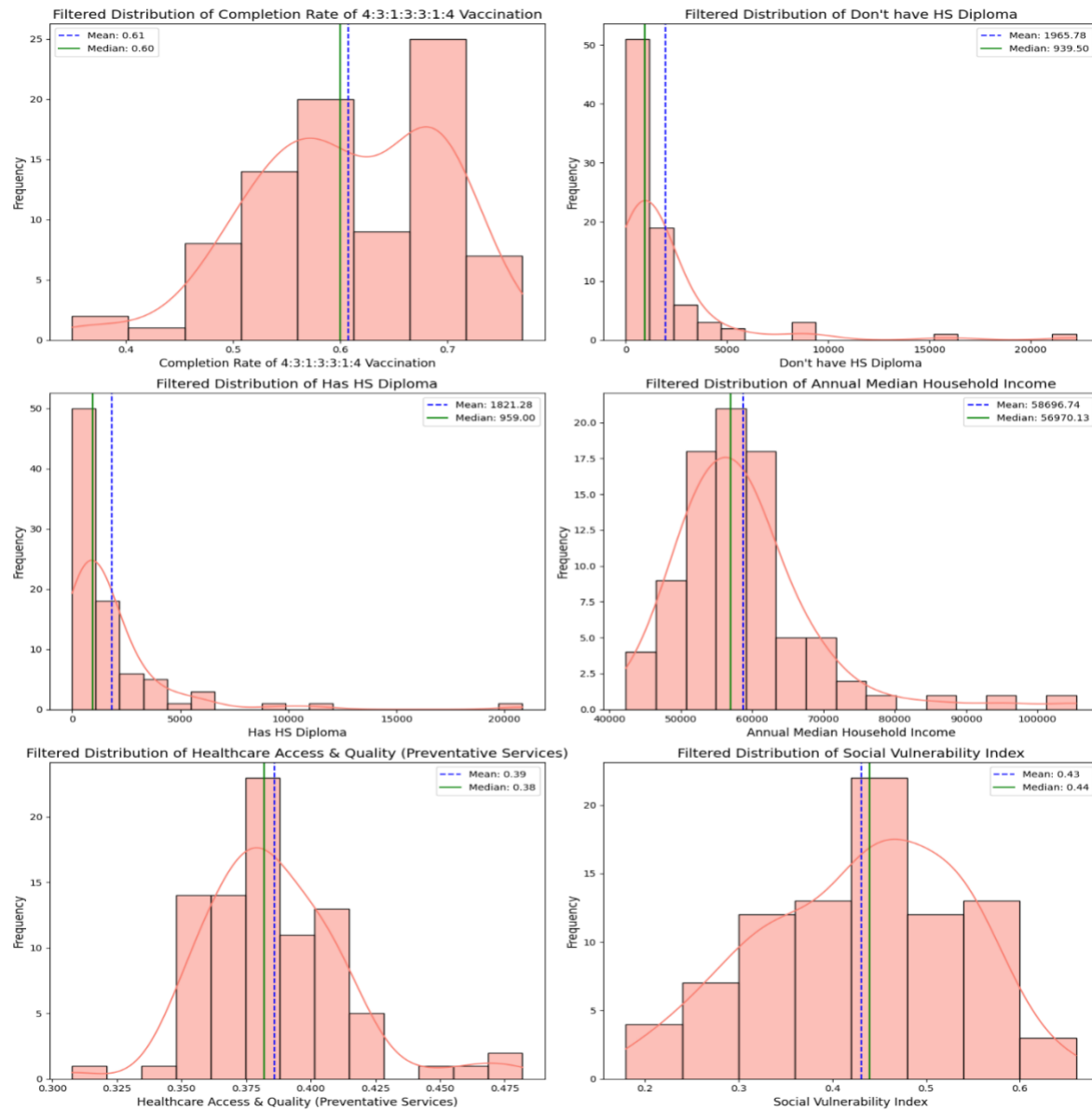


Fig 16. Graphical representation of Skewness and Distribution Trends

```

import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('Baseline.csv')

cols_info = {
    'COMPLETION_RATE': "Completion Rate of 4:3:1:3:3:1:4 Vaccination",
    'NO_HS_DIPLOMA_COUNT': "Don't have HS Diploma",
    'HS_DIPLOMA_COUNT': "Has HS Diploma",
    'AVG_MED_INCM': "Annual Median Household Income",
    'AVG_PREV': "Healthcare Access & Quality (Preventative Services)",
    'AVG_SVI': "Social Vulnerability Index"
}

df_filtered = df[(df['AVG_SVI'] >= 0) & (df['AVG_SVI'] <= 1)].copy()

stats_results = []
for col, label in cols_info.items():
    data = df_filtered[col].dropna()

    mean_val = data.mean()
    median_val = data.median()
    mode_series = data.mode()
    mode_val = mode_series.iloc[0] if not mode_series.empty else "N/A"

    stats_results.append({
        "Variable": label,
        "Mean": round(mean_val, 4),
        "Median": round(median_val, 4),
        "Mode": round(mode_val, 4) if isinstance(mode_val, (int, float)) else mode_val
    })

stats_df = pd.DataFrame(stats_results)

fig, axes = plt.subplots(nrows=3, ncols=2, figsize=(15, 18))
axes = axes.flatten()

for i, (col, label) in enumerate(cols_info.items()):
    sns.histplot(df_filtered[col], kde=True, ax=axes[i], color='salmon')
    axes[i].set_title(f'Filtered Distribution of {label}', fontsize=14)
    axes[i].set_xlabel(label, fontsize=12)
    axes[i].set_ylabel('Frequency', fontsize=12)

    m_val = df_filtered[col].mean()
    med_val = df_filtered[col].median()
    axes[i].axvline(m_val, color='blue', linestyle='--', label=f'Mean: {m_val:.2f}')
    axes[i].axvline(med_val, color='green', linestyle='--', label=f'Median: {med_val:.2f}')
    axes[i].legend()

plt.tight_layout()
plt.savefig('filtered_variable_distributions.png')

print("Descriptive Statistics (Filtered: 0 <= AVG_SVI <= 1):")
print(stats_df.to_string(index=False))

print(f"\nOriginal row count: {len(df)}")
print(f"Filtered row count: {len(df_filtered)}")

```

Fig 17. Normality Testing and Graphical Representation

5.3 Testing the correlation of each independent variable to the dependent variable

Based on the statistical analysis performed on the Completion Rate of the 4:3:1:3:3:1:4 Vaccination series, we examined the relationship between several socioeconomic factors and vaccination outcomes. Because the data previously showed non-normal distributions, we utilized non-parametric tests, specifically the Spearman Correlation for continuous variables and the Mann-Whitney U Test for categorical comparisons.

Determinant	Test Used	P value	Correlation Value
No HS Diploma	Spearman Correlation	0.7712	-0.0307
Has HS Diploma or Above	Spearman Correlation	0.6403	0.0494
Average Annual Median Household Income	Spearman Correlation	0.6719	0.0447
Healthcare Access & Quality (Preventative Services)	Spearman Correlation	0.0079	0.2753
Social Vulnerability Index	Spearman Correlation	0.0064	-0.2823
Location	Mann-Whitney U Test	0.7276	1011.00 (U Statistic)

Table 4. Results of Correlation Analysis

The following sections elaborate on the findings summarized in the provided table:

Significant Socioeconomic Determinants

Two variables demonstrated a statistically significant relationship with vaccination completion rates (where $P < 0.05$):

- **Social Vulnerability Index (SVI):** This factor showed a significant negative correlation ($r = -0.2823$, $P = 0.0064$). This suggests that as social vulnerability increases (indicating higher levels of poverty, lack of transportation, or crowded housing), the vaccination completion rate tends to decrease.
- **Healthcare Access & Quality (Preventative Services):** This determinant showed a significant positive correlation ($r = 0.2753$, $P = 0.0079$). This indicates that counties with

better access to and higher quality of preventative healthcare services typically see higher vaccination completion rates.

Non-Significant Socioeconomic Factors

Several variables showed weak correlations and high P-values, suggesting that, within this specific dataset, they were not primary drivers of the vaccination rate:

- **Educational Attainment:** Both "No HS Diploma" ($P = 0.7712$) and "Has HS Diploma or Above" ($P = 0.6403$) yielded very low correlation values (-0.0307 and 0.0494 , respectively). This indicates that education level alone was not a strong predictor of series completion in this group.
- **Average Annual Median Household Income:** Similar to education, income levels did not show a significant relationship ($P = 0.6719$, $r = 0.0447$), suggesting that income disparity was not the leading barrier to vaccination in the analyzed counties.
- **Geographic Location:** The Mann-Whitney U Test was used to compare Rural vs. Urban locations. With a P value of 0.7276 and a U Statistic of 1011.00 , the results suggest there is no statistically significant difference in vaccination completion rates based solely on whether a county is classified as rural or urban.

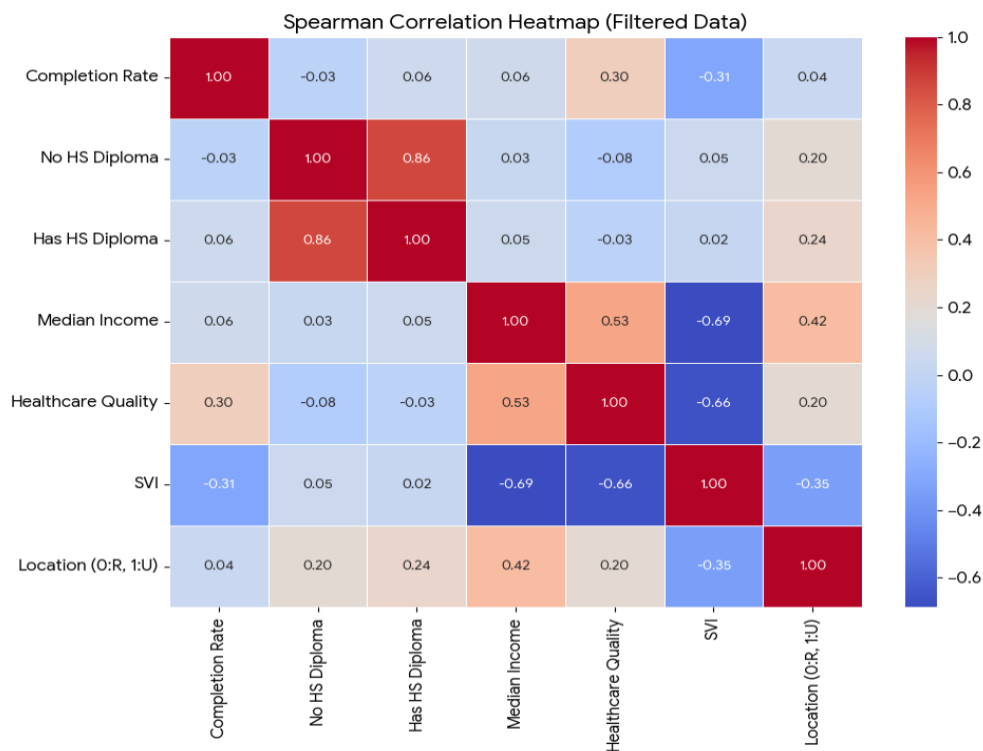


Fig 18. Correlation Heatmap

```
import pandas as pd
from scipy.stats import spearmanr

df = pd.read_csv('Baseline.csv')

output_col = 'COMPLETION_RATE'
input_cols = [
    'NO_HS_DIPLOMA_COUNT',
    'HS_DIPLOMA_COUNT',
    'AVG_MED_INCM',
    'AVG_PREV',
    'AVG_SVI'
]

results = []

for col in input_cols:

    temp_df = df[[output_col, col]].dropna()

    correlation, p_value = spearmanr(temp_df[output_col], temp_df[col])

    results.append({
        'Variable': col,
        'Spearman Correlation': round(correlation, 4),
        'P-Value': round(p_value, 4)
    })

results_df = pd.DataFrame(results)
print("Spearman Correlation Analysis with", output_col)
print("=====")
print(results_df.to_string(index=False))

print("\nInterpretation:")
for index, row in results_df.iterrows():
    significance = "Significant" if row['P-Value'] < 0.05 else "Not Significant"
    print(f"- {row['Variable']}: {significance} (p = {row['P-Value']})")
```

Fig 19. Code for Spearman Correlation

Spearman Correlation Analysis with COMPLETION_RATE

Variable	Spearman Correlation	P-Value
NO_HS_DIPLOMA_COUNT	-0.0307	0.7712
HS_DIPLOMA_COUNT	0.0494	0.6403
AVG_MED_INCM	0.0447	0.6719
AVG_PREV	0.2753	0.0079
AVG_SVI	-0.2823	0.0064

Interpretation:

- NO_HS_DIPLOMA_COUNT: Not Significant (p = 0.7712)
- HS_DIPLOMA_COUNT: Not Significant (p = 0.6403)
- AVG_MED_INCM: Not Significant (p = 0.6719)
- AVG_PREV: Significant (p = 0.0079)
- AVG_SVI: Significant (p = 0.0064)

Fig 20. Results of Spearman Correlation

```
import pandas as pd
from scipy.stats import mannwhitneyu

df = pd.read_csv('Baseline.csv')

cat_col = 'LOCATION'
num_col = 'COMPLETION_RATE'

rural_group = df[df[cat_col] == 'Rural'][num_col].dropna()
urban_group = df[df[cat_col] == 'Urban'][num_col].dropna()

stat, p_value = mannwhitneyu(rural_group, urban_group)

summary_stats = df.groupby(cat_col)[num_col].agg(['mean', 'median', 'count']).reset_index()

print("Descriptive Statistics by Location")
print("=====")
print(summary_stats.to_string(index=False))

print("\nStatistical Test Results")
print("=====")
print(f"Test Performed: Mann-Whitney U test")
print(f"U-Statistic: {stat:.4f}")
print(f"P-Value: {p_value:.4f}")

significant = "Yes" if p_value < 0.05 else "No"
higher_group = "Urban" if summary_stats.loc[1, 'median'] > summary_stats.loc[0, 'median'] else "Rural"

print(f"\nSummary Table:")
results_table = pd.DataFrame([
    "Comparison": "Rural vs Urban",
    "P-Value": round(p_value, 4),
    "Significant?": significant,
    "Higher Completion Rate": higher_group if significant == "Yes" else "No Significant Difference"
])
print(results_table.to_string(index=False))
```

Fig 21. Code for Mann-Whitney U Test

Descriptive Statistics by Location

LOCATION	mean	median	count
Rural	0.601875	0.600	48
Urban	0.614318	0.625	44

Statistical Test Results

Test Performed: Mann-Whitney U test
U-Statistic: 1011.0000
P-Value: 0.7276

Summary Table:

Comparison	P-Value	Significant?	Higher Completion Rate
Rural vs Urban	0.7276	No	No Significant Difference

Fig 22. Results of Mann-Whitney U Test

5.4 Additional Steps

In this comparative analysis, we evaluated the efficacy of linear and logistic regression models utilizing a dataset primarily composed of continuous numerical variables. To satisfy the requirements of the logistic regression framework, specific continuous predictors were discretized into categorical formats through Python-based quartile partitioning, a process conducted in accordance with rigorous statistical methodologies [3, 9]. Notably, variables such as the Social Vulnerability Index and Median Household Income remained in their original form, as these metrics are pre-standardized into categorical indices by governing regulatory bodies and established research organizations, thereby precluding the need for further manual transformation [2, 4].

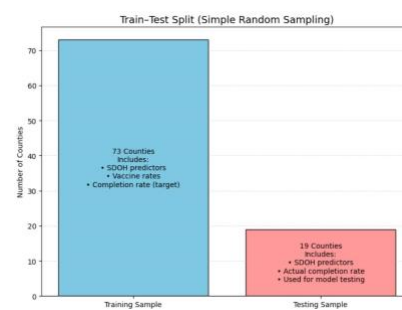
6 Machine Learning Models

6.1 Data Preparation for Machine Learning

Fig 23. SDOH data separated into bins

COUNTY	COMPLETION_RATE	NO_HS_EDU_BIN	HS_EDU_BIN	SVI_CAT	PREV_BIN	INC_CAT
Adams	0.5	Q1	Q2	More Vulnerable	Q2	Lower Income
Allen	0.58	Q4	Q4	Out of Range	Q1	Middle Income
Bartholom	0.54	Q3	Q4	Less Vulnerable	Q4	Middle Income
Benton	0.51	Q1	Q1	More Vulnerable	Q1	Lower Income
Blackford	0.59	Q1	Q1	More Vulnerable	Q2	Lower Income
Boone	0.71	Q3	Q3	Least Vulnerable	Q4	Middle Income
Brown	0.63	Q1	Q1	Less Vulnerable	Q4	Middle Income
Carroll	0.6	Q1	Q1	Less Vulnerable	Q4	Lower Income
Cass	0.57	Q3	Q2	More Vulnerable	Q1	Lower Income
Clark	0.56	Q4	Q4	Less Vulnerable	Q1	Middle Income
Clay	0.66	Q3	Q3	Less Vulnerable	Q3	Middle Income

Fig 24. Train-Test Split



Our team prepared the data by categorizing continuous variables into quartiles to capture meaningful patterns and simplify interpretation. We then applied a train-test split to create separate subsets for model training and evaluation, which helped increase variability and prevent overfitting before running the machine learning models.

6.2 Logistic Regression Results

We used logistic regression to classify Indiana counties into high versus low childhood vaccination completion based on binned county-level SDOH indicators. Model performance was evaluated using a confusion matrix, ROC curve, precision-recall curve, and a classification report to ensure a comprehensive assessment of classification quality.

Fig 25. Python code and results for Logistic Regression Classification Report

The classification report indicates limited predictive performance. Overall accuracy was 0.54, only slightly better than random chance. For counties classified as *high completion*, the model achieved a precision of 0.58 and recall of 0.69, suggesting it was moderately effective at identifying higher-performing counties but did so at the cost of misclassifying lower-performing ones. Performance for *low completion* counties was weaker, with precision of 0.44 and recall of 0.33, indicating difficulty correctly identifying counties with poorer vaccination outcomes.

```
log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_scaled, y_train_bin)

y_pred_log = log_reg.predict(X_test_scaled)

print("LOGISTIC REGRESSION CLASSIFICATION REPORT")
print(classification_report(y_test_bin, y_pred_log))
```

LOGISTIC REGRESSION CLASSIFICATION REPORT					
	precision	recall	f1-score	support	
0	0.44	0.33	0.38	12	
1	0.58	0.69	0.63	16	
accuracy			0.54	28	
macro avg	0.51	0.51	0.50	28	
weighted avg	0.52	0.54	0.52	28	

Fig 26. Python Code and Results for Logistic Regression Confusion Matrix

```
cm = confusion_matrix(y_test_bin, y_pred)

disp = ConfusionMatrixDisplay(confusion_matrix=cm)
disp.plot()
plt.title("Confusion Matrix - Logistic Regression")
plt.show()
```

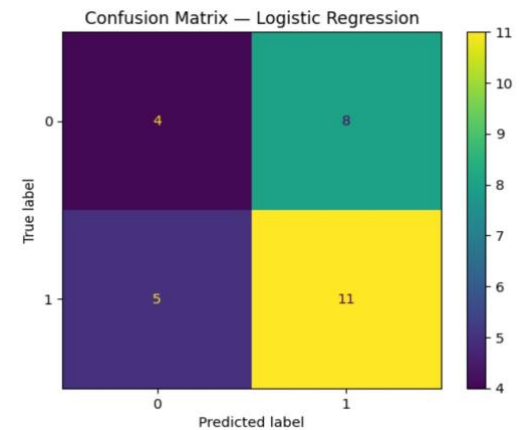


Fig 27. Python Code and Results for Logistic Regression ROC Curve

```
fpr, tpr, _ = roc_curve(y_test_bin, y_prob)
roc_auc = auc(fpr, tpr)

plt.figure()
plt.plot(fpr, tpr, label=f"ROC AUC = {roc_auc:.2f}")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve - Logistic Regression")
plt.legend()
plt.show()
```

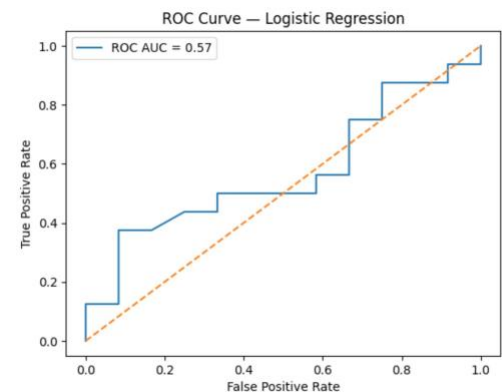
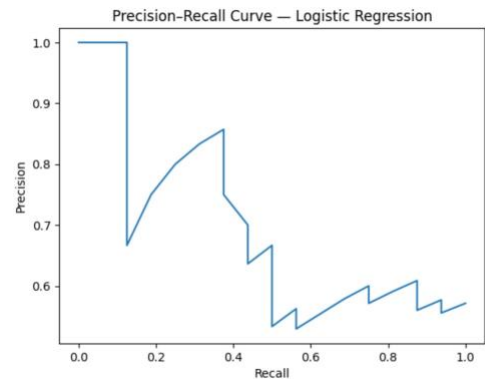


Fig 28. Python Code and Results for Logistic Regression Precision-Recall Curve

```
precision, recall, _ = precision_recall_curve(y_test_bin, y_prob)

plt.figure()
plt.plot(recall, precision)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve - Logistic Regression")
plt.show()
```



The ROC AUC of 0.57 further confirms weak discriminatory ability. The ROC curve lies close to the diagonal reference line, demonstrating that the model only marginally improves upon random classification. Similarly, the precision–recall curve shows limited separation between positive and negative classes, reflecting unstable precision across recall levels. The confusion matrix reinforces these findings, showing a notable number of false positives and false negatives. Overall, our logistic regression results suggest that broad county-level SDOH bins alone are insufficient for reliably classifying vaccination completion status.

6.3 Linear Regression Results

We also evaluated a linear regression model, first predicting continuous vaccination completion rates and then converting predictions into binary classes using a median threshold. As with logistic regression, we assessed performance using a confusion matrix, ROC curve, precision–recall curve, and classification report.

Fig 29. Python Code and Results for Linear Regression Classification Report

The linear regression model demonstrated moderately improved performance relative to logistic regression. The classification report shows an overall accuracy of 0.57, with balanced precision and recall across both classes. For *high completion* counties, precision and recall were both 0.62, indicating more consistent identification compared to logistic regression. Performance for *low completion* counties was also balanced, with precision and recall each at 0.50, reflecting fewer extreme misclassifications.

```
from sklearn.metrics import classification_report

print("LINEAR REGRESSION - CLASSIFICATION REPORT")
print(classification_report(y_test_bin, y_pred_lin_bin))
```

LINEAR REGRESSION - CLASSIFICATION REPORT				
	precision	recall	f1-score	support
0	0.50	0.50	0.50	12
1	0.62	0.62	0.62	16
accuracy			0.57	28
macro avg	0.56	0.56	0.56	28
weighted avg	0.57	0.57	0.57	28

Fig 30. Python Code and Results for Linear Regression Confusion Matrix

```
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm_lin = confusion_matrix(y_test_bin, y_pred_lin_bin)

disp_lin = ConfusionMatrixDisplay(confusion_matrix=cm_lin)
disp_lin.plot()
plt.title("Confusion Matrix – Linear Regression (Thresholded)")
plt.show()
```

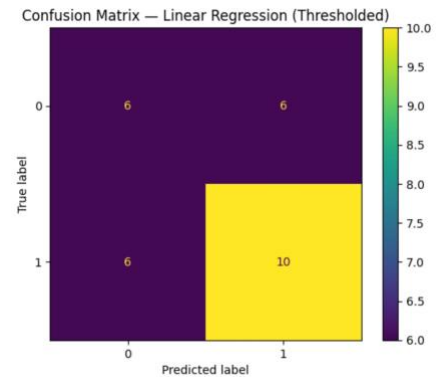


Fig 31. Python Code and Results for Linear Regression ROC Curve

```
from sklearn.metrics import roc_curve, auc

fpr_lin, tpr_lin, _ = roc_curve(y_test_bin, y_pred_lin_cont)
roc_auc_lin = auc(fpr_lin, tpr_lin)

plt.figure()
plt.plot(fpr_lin, tpr_lin, label=f"Linear Reg ROC AUC = {roc_auc_lin:.2f}")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve – Linear Regression (Thresholded)")
plt.legend()
plt.show()
```

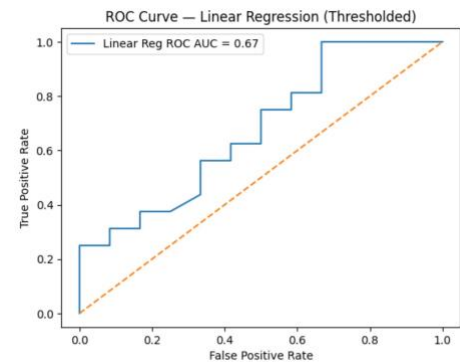
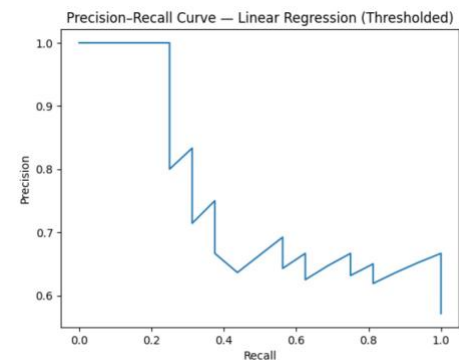


Fig 32. Python Code and Results for Linear Regression Precision-Recall Curve

```
from sklearn.metrics import precision_recall_curve

precision_lin, recall_lin, _ = precision_recall_curve(
    y_test_bin, y_pred_lin_cont
)

plt.figure()
plt.plot(recall_lin, precision_lin)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision-Recall Curve – Linear Regression (Thresholded)")
plt.show()
```



The ROC AUC of 0.67 indicates moderate discriminative ability, suggesting that the continuous predictions from linear regression capture meaningful signals in the data when thresholded for classification. The ROC curve shows clearer separation from the diagonal compared to logistic regression, and the precision–recall curve demonstrates improved stability across recall values. The confusion matrix supports these findings, showing a more even distribution of correct predictions across both outcome classes. Despite this improvement, misclassification remains substantial, highlighting that even the stronger linear regression approach struggles when relying solely on aggregated county-level SDOH indicators.

7 Summary of Findings

Our analysis combined correlation testing, statistical modeling, and machine learning approaches to evaluate how county-level social determinants of health (SDOH) relate to childhood vaccination completion rates across Indiana. Taken together, our findings support the null hypothesis that education alone is not a strong predictor of childhood vaccination completion at the county level.

Spearman correlation analysis revealed statistically significant associations between vaccination completion rates and two SDOH indicators: preventative service utilization (Healthcare Access & Quality) and the Social Vulnerability Index (SVI). Preventative services showed a Spearman p -value of 0.0079, and SVI showed a p -value of 0.0064, indicating meaningful monotonic relationships with vaccination outcomes. These results suggest that counties with better access to preventive care and lower social vulnerability tend to exhibit higher vaccination completion rates. However, correlation alone does not imply predictive capability.

When we evaluated these same SDOH variables using machine learning models, their predictive limitations became clear. Logistic regression produced weak classification performance (ROC AUC = 0.57), indicating minimal ability to distinguish between high- and low-performing counties. Linear regression, when used as a classifier, performed moderately better (ROC AUC = 0.67), but still misclassified a substantial number of counties. These results demonstrate that although certain SDOH variables are statistically associated with vaccination outcomes, they do not provide sufficient signal for accurate prediction when used in aggregated, county-level form.

Overall, our findings highlight a critical distinction in health informatics between statistical significance and predictive utility. While SVI and preventative service utilization show meaningful associations with vaccination completion, county-level SDOH indicators alone do not adequately capture the complex, multi-factor dynamics that drive immunization behavior. Vaccination completion is likely influenced by more granular factors such as vaccine-specific uptake patterns, healthcare access logistics, community trust, and local public health infrastructure. These results reinforce the importance of integrating correlation analysis with predictive modeling and underscore the need for richer, more detailed datasets to inform data-driven public health decision-making.

8 Limitations

Several limitations should be considered when interpreting our findings. Our analysis relied on aggregated county-level data, which may obscure meaningful variation within counties. Vaccination behavior is influenced by individual-, provider-, and neighborhood-level factors that cannot be fully captured through broad SDOH indicators. While we attempted to obtain more granular data, such as clinic-level access, appointment availability, or household-level education, recent datasets (within the past five years) were not publicly available for Indiana.

The limited sample size of approximately 92 counties constrained statistical power and restricted the complexity of machine learning models that could be reliably applied. This limitation likely

contributed to the modest predictive performance observed, particularly for classification tasks. Additionally, our study used cross-sectional data, which limits the ability to draw causal conclusions. Although Spearman correlation analysis identified statistically significant associations between vaccination completion and variables such as social vulnerability and preventive service utilization, these findings reflect correlation rather than causation.

Lastly, the predictive models were restricted to the variables included in the dataset. Influences such as vaccine-specific access barriers, provider availability, public health outreach efforts, community trust, and religious or cultural factors were not measurable and may play a critical role in vaccination outcomes. Despite these constraints, the project highlights important gaps in available data and provides a foundation for more targeted, data-driven public health research.

APPENDIX

Appendix A: SQL Code for Importing Excel Data into MySQL

Step 1: Convert Excel to CSV (Pre-SQL Step)

- Open the Excel file
- Click **File** → **Save As**
- Select **CSV (Comma delimited) (.csv)**
- Save the file (example: master_sdoh.csv)

This step ensures compatibility with MySQL.

Step 2: Create Database

```
CREATE DATABASE sdoh_db;  
USE sdoh_db;
```

Step 3: Create Table Structure

```
CREATE TABLE master_sdoh (  
  County VARCHAR(100),  
  County_ID INT,  
  No_Formal_Ed_Count INT,  
  No_HS_Diploma_Count INT,  
  HS_Diploma_Count INT,  
  Associates_Count INT,  
  Bachelors_Count INT,  
  Grad_School_Count INT,  
  Attending_School_Count INT,  
  Completion_Rate DECIMAL(5,3),  
  Dtap_Rate DECIMAL(5,3),
```

```

Polio_Rate DECIMAL(5,3),
MMR_Rate DECIMAL(5,3),
Hib_Rate DECIMAL(5,3),
HepB_Rate DECIMAL(5,3),
Var_Rate DECIMAL(5,3),
PCV_Rate DECIMAL(5,3),
Metro TINYINT,
Non_Metro TINYINT,
Avg_Median_Household_Income DECIMAL(10,2),
Avg_Current_Preventive_Services_65 DECIMAL(5,2),
SVI_County_Average DECIMAL(5,2));

```

Step 5: Verify Imported Data

```

SELECT * FROM master_sdoh LIMIT 10;
SELECT COUNT(*) FROM master_sdoh;

```

Database Connection and Data Extraction MySQL → Python Database Connection and Data Extraction (MySQL → Python)

```

import mysql.connector
import pandas as pd

```

```

# Connect to MySQL database
conn = mysql.connector.connect(
    host="localhost",
    user="DB_USERNAME",
    password="DB_PASSWORD",
    database="DB_NAME"
)

```

```

# Load data from MySQL into pandas DataFrame
df = pd.read_sql("SELECT * FROM master_sdoh_1_csv;", conn)

```

```

# Preview data
df.head()
df.info()

```

Histogram – Vaccination Completion Rate Distribution

```

import numpy as np
import matplotlib.pyplot as plt

data = df['COMPLETION_RATE'].dropna()

counts, bins, patches = plt.hist(
    data,
    bins=10,
    edgecolor='black'
)

```

```
)

colors = plt.cm.tab10(np.linspace(0, 1, len(patches)))
for patch, color in zip(patches, colors):
    patch.set_facecolor(color)

plt.title("Distribution of Childhood Vaccination Completion Rates")
plt.xlabel("Completion Rate")
plt.ylabel("Frequency")
plt.grid(axis='y', alpha=0.4)
plt.show()
```

Normality Test – Box-Cox Transformation

```
import numpy as np
import matplotlib.pyplot as plt
from scipy import stats

target = df['COMPLETION_RATE'].dropna()

# Box-Cox transformation
transformed_data, lambda_val = stats.boxcox(target)

# Shapiro-Wilk test
stat, p_value = stats.shapiro(transformed_data)

print("Lambda:", lambda_val)
print("Box-Cox p-value:", p_value)

# Histogram
plt.hist(transformed_data, bins=10, density=True, alpha=0.6, edgecolor='black')
x = np.linspace(min(transformed_data), max(transformed_data), 100)
plt.plot(x, stats.norm.pdf(x, np.mean(transformed_data), np.std(transformed_data)), 'r-',
linewidth=2)

plt.title("Normality Check: Vaccination Completion Rate")
plt.xlabel("Completion Rate")
plt.ylabel("Density")
plt.show()
```

Scatter Plot – SDOH Risk Factors vs Completion Rate

```
import matplotlib.pyplot as plt

cols = [
    'NO_FORMAL_ED_COUNT',
    'NO_HS_DIPLOMA_COUNT',
```

```

    'HS_DIPLOMA_COUNT',
    'AVG_MED_INC',
    'SVI County Average'
]

plt.figure(figsize=(10,6))

for col in cols:
    plt.scatter(df[col], df['COMPLETION_RATE'], alpha=0.7, label=col)

plt.xlabel("SDOH Risk Factor Value")
plt.ylabel("Vaccination Completion Rate")
plt.title("Distribution of Key SDOH Risk Factors vs Vaccination Completion Rate")
plt.legend()
plt.grid(alpha=0.3)
plt.show()

```

Scatter Plot with Trendline

```

import numpy as np
import matplotlib.pyplot as plt

x = df['AVG_MED_INC']
y = df['COMPLETION_RATE']

plt.scatter(x, y, alpha=0.7, s=40, color='blue')

# Trendline
m, b = np.polyfit(x, y, 1)
xs = np.linspace(x.min(), x.max(), 100)
plt.plot(xs, m*xs + b, color='red', linewidth=2)

plt.xlabel("Average Median Household Income")
plt.ylabel("Completion Rate")
plt.title("Completion Rate vs Average Median Household Income")
plt.grid(alpha=0.3)
plt.show()

```

Top 10 and Bottom 10 Vaccinated Counties

Column name

```
vacc_col = 'COMPLETION_RATE'
```

Top 10 counties

```
top10 = df.sort_values(by=vacc_col, ascending=False).head(10)

plt.figure(figsize=(8,5)) plt.barh(top10['County'], top10[vacc_col], color='green')
plt.xlabel("Vaccination Completion Rate") plt.title("Top 10 Highest Vaccinated Counties")
plt.gca().invert_yaxis() plt.show()
```

Bottom 10 counties

```
bottom10 = df.sort_values(by=vacc_col, ascending=True).head(10)

plt.figure(figsize=(8,5)) plt.barh(bottom10['County'], bottom10[vacc_col], color='red')
plt.xlabel("Vaccination Completion Rate") plt.title("Bottom 10 Lowest Vaccinated Counties")
plt.gca().invert_yaxis() plt.show()
```

Code For Normality Test of variables, finding Mean, Median, Mode & Graphical Representation of the results

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

df = pd.read_csv('Baseline.csv')

cols_info = {
    'COMPLETION_RATE': "Completion Rate of 4:3:1:3:3:1:4 Vaccination",
    'NO_HS_DIPLOMA_COUNT': "Don't have HS Diploma",
    'HS_DIPLOMA_COUNT': "Has HS Diploma",
    'AVG_MED_INCM': "Annual Median Household Income",
    'AVG_PREV': "Healthcare Access & Quality (Preventative Services)",
    'AVG_SVI': "Social Vulnerability Index"
}

df_filtered = df[(df['AVG_SVI'] >= 0) & (df['AVG_SVI'] <= 1)].copy()

stats_results = []
for col, label in cols_info.items():
    data = df_filtered[col].dropna()

    mean_val = data.mean()
    median_val = data.median()
    mode_series = data.mode()
    mode_val = mode_series.iloc[0] if not mode_series.empty else "N/A"
```

```

stats_results.append({
    "Variable": label,
    "Mean": round(mean_val, 4),
    "Median": round(median_val, 4),
    "Mode": round(mode_val, 4) if isinstance(mode_val, (int, float)) else mode_val
})

stats_df = pd.DataFrame(stats_results)

fig, axes = plt.subplots(nrows=3, ncols=2, figsize=(15, 18))
axes = axes.flatten()

for i, (col, label) in enumerate(cols_info.items()):
    sns.histplot(df_filtered[col], kde=True, ax=axes[i], color='salmon')
    axes[i].set_title(f'Filtered Distribution of {label}', fontsize=14)
    axes[i].set_xlabel(label, fontsize=12)
    axes[i].set_ylabel('Frequency', fontsize=12)

    m_val = df_filtered[col].mean()
    med_val = df_filtered[col].median()
    axes[i].axvline(m_val, color='blue', linestyle='--', label=f'Mean: {m_val:.2f}')
    axes[i].axvline(med_val, color='green', linestyle='-', label=f'Median: {med_val:.2f}')
    axes[i].legend()

plt.tight_layout()
plt.savefig('filtered_variable_distributions.png')

print("Descriptive Statistics (Filtered: 0 <= AVG_SVI <= 1):")
print(stats_df.to_string(index=False))

print(f"\nOriginal row count: {len(df)}")
print(f"Filtered row count: {len(df_filtered)}")

```

Appendix B: Python codes for statistical analysis on Jupyterhub

Code for testing correlation between dependent variable and individual independent variables

For numerical independent variables

```

import pandas as pd
from scipy.stats import spearmanr

df = pd.read_csv('Baseline.csv')

```



```

output_col = 'COMPLETION_RATE'
input_cols = [
    'NO_HS_DIPLOMA_COUNT',
    'HS_DIPLOMA_COUNT',
    'AVG_MED_INCM',
    'AVG_PREV',
    'AVG_SVI'
]

results = []

for col in input_cols:

    temp_df = df[[output_col, col]].dropna()

    correlation, p_value = spearmanr(temp_df[output_col], temp_df[col])

    results.append({
        'Variable': col,
        'Spearman Correlation': round(correlation, 4),
        'P-Value': round(p_value, 4)
    })

results_df = pd.DataFrame(results)
print("Spearman Correlation Analysis with", output_col)
print("=====")
print(results_df.to_string(index=False))

print("\nInterpretation:")
for index, row in results_df.iterrows():
    significance = "Significant" if row['P-Value'] < 0.05 else "Not Significant"
    print(f"- {row['Variable']}: {significance} (p = {row['P-Value']})")

```

For categorical dependent variable

```

import pandas as pd
from scipy.stats import mannwhitneyu

df = pd.read_csv('Baseline.csv')

cat_col = 'LOCATION'
num_col = 'COMPLETION_RATE'

rural_group = df[df[cat_col] == 'Rural'][num_col].dropna()
urban_group = df[df[cat_col] == 'Urban'][num_col].dropna()

```

```

stat, p_value = mannwhitneyu(rural_group, urban_group)

summary_stats = df.groupby(cat_col)[num_col].agg(['mean', 'median', 'count']).reset_index()

print("Descriptive Statistics by Location")
print("=====")
print(summary_stats.to_string(index=False))

print("\nStatistical Test Results")
print("=====")
print(f"Test Performed: Mann-Whitney U test")
print(f"U-Statistic: {stat:.4f}")
print(f"P-Value: {p_value:.4f}")

significant = "Yes" if p_value < 0.05 else "No"
higher_group = "Urban" if summary_stats.loc[1, 'median'] > summary_stats.loc[0, 'median'] else
"Rural"

print(f"\nSummary Table:")
results_table = pd.DataFrame([
    "Comparison": "Rural vs Urban",
    "P-Value": round(p_value, 4),
    "Significant?": significant,
    "Higher Completion Rate": higher_group if significant == "Yes" else "No Significant
Difference"
])
print(results_table.to_string(index=False))

```

Appendix C: Python Code for Logistic and Linear Regression

Logistic Regression

```

log_reg = LogisticRegression(max_iter=1000)
log_reg.fit(X_train_scaled, y_train_bin)

y_pred_log = log_reg.predict(X_test_scaled)

lin_reg = LinearRegression()
lin_reg.fit(X_train_scaled, y_train_cont)

y_pred_lin = lin_reg.predict(X_test_scaled)

print("LINEAR REGRESSION RESULTS")
print(f"R2: {r2_score(y_test_cont, y_pred_lin):.3f}")
print(f"MAE: {mean_absolute_error(y_test_cont, y_pred_lin):.3f}")

```

```
print(f'RMSE: {np.sqrt(mean_squared_error(y_test_cont, y_pred_lin)):.3f}')
```

```
lin_coef_df = pd.DataFrame({  
    "Feature": X.columns,  
    "Coefficient": lin_reg.coef_  
}).sort_values(by="Coefficient", key=abs, ascending=False)
```

```
lin_coef_df
```

```
fpr, tpr, _ = roc_curve(y_test_bin, y_prob)  
roc_auc = auc(fpr, tpr)
```

```
plt.figure()  
plt.plot(fpr, tpr, label=f'ROC AUC = {roc_auc:.2f}')  
plt.plot([0, 1], [0, 1], linestyle="--")  
plt.xlabel("False Positive Rate")  
plt.ylabel("True Positive Rate")  
plt.title("ROC Curve — Logistic Regression")  
plt.legend()  
plt.show()
```

```
precision, recall, _ = precision_recall_curve(y_test_bin, y_prob)
```

```
plt.figure()  
plt.plot(recall, precision)  
plt.xlabel("Recall")  
plt.ylabel("Precision")  
plt.title("Precision–Recall Curve — Logistic Regression")  
plt.show()
```

```
cm = confusion_matrix(y_test_bin, y_pred)
```

```
disp = ConfusionMatrixDisplay(confusion_matrix=cm)  
disp.plot()  
plt.title("Confusion Matrix — Logistic Regression")  
plt.show()
```

```
log_coef_df = pd.DataFrame({  
    "Feature": X.columns,  
    "Coefficient": log_reg.coef_[0]  
}).sort_values(by="Coefficient", key=abs, ascending=False)
```

```
log_coef_df
```

Linear Regression

```
lin_reg = LinearRegression()
lin_reg.fit(X_train_scaled, y_train_cont)

y_pred_lin_cont = lin_reg.predict(X_test_scaled)

threshold = median_threshold

y_pred_lin_bin = (y_pred_lin_cont >= threshold).astype(int)

from sklearn.metrics import roc_curve, auc

fpr_lin, tpr_lin, _ = roc_curve(y_test_bin, y_pred_lin_cont)
roc_auc_lin = auc(fpr_lin, tpr_lin)

plt.figure()
plt.plot(fpr_lin, tpr_lin, label=f"Linear Reg ROC AUC = {roc_auc_lin:.2f}")
plt.plot([0, 1], [0, 1], linestyle="--")
plt.xlabel("False Positive Rate")
plt.ylabel("True Positive Rate")
plt.title("ROC Curve — Linear Regression (Thresholded)")
plt.legend()
plt.show()

from sklearn.metrics import precision_recall_curve

precision_lin, recall_lin, _ = precision_recall_curve(
    y_test_bin, y_pred_lin_cont
)

plt.figure()
plt.plot(recall_lin, precision_lin)
plt.xlabel("Recall")
plt.ylabel("Precision")
plt.title("Precision–Recall Curve — Linear Regression (Thresholded)")
plt.show()

from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

cm_lin = confusion_matrix(y_test_bin, y_pred_lin_bin)

disp_lin = ConfusionMatrixDisplay(confusion_matrix=cm_lin)
disp_lin.plot()
plt.title("Confusion Matrix — Linear Regression (Thresholded)")
plt.show()
```

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